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ZE-194

B.Tech. (New Scheme) (B.E.)

(Main/ATKT) Examination, 2026

(Fourth Semester)

ANALOG AND DIGITAL COMMUNICATION

EL-403

Time : 3 Hours]

[Maximum Marks : 60

Note : Attempt all questions. All questions carry equal marks.

1. A 100 MHz carrier is frequency modulated by a 5 kHz sinusoidal signal. If the maximum frequency deviation is 50 kHz, calculate :

- (i) The modulation index
- (ii) The bandwidth of the FM signal using Carson's rule
- (iii) If the amplitude of the modulating signal is doubled, how does the bandwidth change ?

Or

- (a) What is the need for modulation ? Explain the Classification of Modulation.
 - (b) Compare the noise performance of AM and FM systems.
- 2.
- (a) Discuss generation and detection of PAM signals. Derive BW and S/N ration of PAM.
 - (b) Discuss Coherent and non-coherent detection techniques.

Or

- (a) What is sampling theorem ? State and prove sampling for band pass filter.
 - (b) Difference between Natural sampling and Flat top sampling.
- 3.
- (a) With a neat diagram and necessary equations explain the working principle of pulse amplitude modulation.

- (b) What is the difference between PAM and PWM ?
How does PAM work ?

Or

- (a) What is pulse modulation ? Explain PCM transmitter receiver along with block diagram.
- (b) Compare the performance of DM and ADM systems.

4. (a) Write a short note on the following :

(i) Binary symmetric channel

(i) Cascaded channel

- (b) Apply the Shannon fano procedure for the following message ensemble :

$$[X] = [x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6 \ x_7 \ x_8]$$

$$[P] = [0.1 \ 0.25 \ 0.15 \ 0.05 \ 0.15 \ 0.1 \ 0.05 \ 0.15]$$

Or

- (a) Explain joint entropy and conditional entropy

- (b) Construct the Huffman code with minimum code variance for the following probabilities and also determine the code variance and code efficiency :
- {0.25, 0.25, 0.125, 0.125, 0.125, 0.0625, 0.0625}
5. (a) Draw the block diagrams that represents the following :
- (i) QPSK Transmitter
 - (ii) QPSK Receiver
- (b) Draw the block diagram of BFSK detect.

Or

- (a) With a block diagram and necessary equations explain generation and detection of DPSK.
- (b) With necessary equations and waveforms, explain the concept of QPSK.

